

LECTURE 5 & 7

***The mechanics of menu planning
using the food guide pyramid**

1-Food guide pyramid

2-Food groups (exchange list)

LECTURE 5

LECTURE 5

Food groups and serving size

The mechanics of menu planning using the food guide pyramid

The mechanics of menu planning

- 1-Food guide pyramid**
- 2-Food groups (exchange list)**
- 3-Food composition table**
- 4-Food Labeling**
- 5-Healthy plate (my plate)**
- 6-Healthy food palm**

FOOD GROUPS

- **Objectives of food groups**
- 1-Classify food items into different food groups;
- 2-State the major nutrients and group the food items with similar nutrients for food exchange.

Food groups

Definition of food groups

A food group consists of a number of food items sharing some common characteristics.

Classification of food groups:

The two ways of classifying food into groups
1-physiological, on the basis of function
2-on the basis of nutrients

Food groups

1-Classification Based on Physiological Functions

Food has three basic physiological functions

A-Energy giving

B-Repair and growth

C-Protection and regulation.

Based on the functions, they can be divided into three groups:

Function	Nutrient	Food
Energy giving food	carbohydrates and fats	cereals, sugar , oil
Body building food	proteins	pulses, milk, meat, chicken
Regulatory and protective foods	vitamins and minerals	fruits and vegetables

2-Classification Based on Nutrient

The classification based on the nutrients which they supply.

Five Food Group System

Food group	Foods	Major nutrients present
1- Cereals, Grains and their Products	Rice, wheat, barley, rice maize, flakes, wheat flour etc.	carbohydrates, vitamin B1, vitamin B2, proteins, iron, fiber and folic acid

2-Classification Based on Nutrient

The classification based on the nutrients which they supply.

FOOD GROUPS	FOODS	Major nutrients present
2-pulses and legumes	Bengal gram, black gram, green gram, red gram lentil ,peas, soya bean	carbohydrates, protein, vitamin B, iron, fiber
3-Milk	Milk, curd, paneer, cheese, buttermilk, skimmed milk.	proteins, fat, vitamin B2, calcium, vitamin A
4-Meat and meat products	Meat – chicken, liver, fish, egg	protein, fat, vitamin A, vitamin B2 and vitamin B12

FOOD GROUPS	FOODS	Major nutrients present
5-Fruits and vegetables	-Mango, guava, orange, papaya, banana, sweet lime, apple . -Vegetables ,pumpkin, Tomatoes and green leafy vegetables like spinach, coriander leaves.	Vitamin A , vitamin C ,carbohydrates and fiber -Carotenoids, vitamin B, folic acid, iron and fiber
6-Fats and sugar	-Fats: butter, ghee hydrogenated oil, ground nut, mustard, coconut oil. -Sugars sugar, jaggery, honey	-Fats Carbohydrates

CHARACTERISTICS OF FOOD GROUPS

1 -Cereals

1-Whole cereals and grains contain considerable amount of fiber that **has a great importance in** maintaining a healthy **digestive system.**

2-Cereals are also rich in vitamin B complex. This gets **lost** when cereals are refined, So our diet should have whole grains.

3-Eat at least 3 ounces of whole grain breads, .cereal, crackers, rice, or pasta each day

3- Milk

1- it is a rich source of protein, fat, vitamin A and calcium.

2-Get low-fat or fat-free dairy products .

3-People who cannot consume dairy can choose lower lactose or lactose-free dairy products or other calcium sources like (Calcium-fortified juices and soy and rice beverages

2- Meat and meat products

A-(Include meat, poultry, fish)

1-Encourage low-fat or lean meats and poultry

2- Cooking methods: baking, broiling, grilling

3- A rich source of high quality **protein, vitamin A**
and vitamin B.

2- Meat and meat products

B – Eggs

1- Eggs are a rich source of almost all nutrients except vitamin C.

2- Eggs contain proteins of excellent quality therefore they are specifically recommended for growing children , pregnant women and lactating mothers.

3- Pulses and legumes

- 1- major source of protein especially for vegetarians.
- 2- They have a appreciable amount of vitamin B, calcium and iron.
- 3- protein quality of pulses can be improved by combining them with cereals.

4-Fruits and Vegetables

A-Fruits:

- 1-Citrus fruits such as oranges and lemons are rich in vit. C.
- 2- Yellow fruits like mango, papaya **contain carotene which is** converted into vitamin A in the body.
- 3-Banana is a good source of iron, calcium **and carbohydrate**
- 4-Pomegranate is rich in iron.
- 5- All fruits contain plenty of fiber, **which helps in bowel** movement.
- 6- Dried fruits like raisins, figs, walnuts and almonds are rich in fats, iron, calcium and fiber.

4-Fruits and Vegetables

B-Vegetables :

Green leafy vegetables: Green leafy vegetables such as spinach, red and green leaf and romaine **and mint are rich in** (carotene, calcium, iron and vitamin C).

5 -Fats and sugar

(Butter, ghee, oils like mustard oil, groundnut oil, coconut oil and soya oil).

1- Encourage people to select health-promoting forms of fats:
fat from fish, nuts, and vegetable oils

2-Limit solid fats: butter, stick margarine, shortening, lard,
and visible fat on meat

3-Some amount of fat is needed in daily diet because they supply
essential fatty acids.

Besides this, some vitamins like

A, D, E and K are fat soluble and important for our body.

FOOD EXCHANGE

The similar food items have been placed together in one food group.

Therefore, if we substitute **one food for the other in the same group** , we will be able to get almost **the same nutrients**.

Meat group

Protein group



Meat



Egg



Fish

Meat group

Food	Serving size
Meats	1 ounce cooked lean beef 1 ounce cooked lean pork or ham
Poultry	1 ounce cooked chicken or turkey, without skin 1 sandwich slice of turkey
Seafood	1 ounce cooked fish or shell fish
Eggs	1 egg

Food	Serving size
Nuts and seeds	1/2 ounce of nuts (12 almonds, 24 pistachios, 7 walnut halves) 1/2 ounce of seeds (pumpkin, sunflower, or squash seeds, hulled, roasted) 1 Tablespoon of peanut butter or almond butter
Beans and peas	1/4 cup of cooked beans (such as black, kidney, pinto, or white beans) 1/4 cup of cooked peas (such as chickpeas,, lentils, or split peas) 1/4 cup roasted soybeans 1 falafel patty (2 1/4", 4 oz) 2 Tablespoons hummus

Milk group

Milk group



Milk



Cheese



Curd

Food	<u>Milk group</u>	Serving size
Milk (choose fat-free or low-fat milk)		1 cup milk
		½ cup evaporated milk
Yogurt (choose fat-free or low-fat yogurt)		1 regular container (8 fluid ounces)
		1 cup yogurt
Cheese (choose reduced-fat or low-fat cheeses)		1 ½ ounces hard cheese (cheddar, mozzarella, Swiss, Parmesan)
		2 ounces processed cheese (American)
		½ cup ricotta cheese
Milk-based desserts (choose fat-free or low-fat types)		1 cup pudding made with milk
		1 cup frozen yogurt
Soy milk (soy beverage)		1 cup calcium-fortified soy milk
		1 half-pint container calcium-fortified soy milk

Vegetables and fruits group

Vegetable and fruit group



Peas



Potatoes



Bananas



Tomatoes

Food	Serving size
Apple	½ large (3 ¼" diameter) 1 small (2 ¼" diameter) 1 cup, sliced or chopped, raw or cooked
Applesauce	1 cup
Banana	1 cup, sliced 1 large (8" to 9" long)
Cantaloupe	1 cup, diced or melon balls
Grapefruit	1 medium (4" diameter) 1 cup, sections
Mixed fruit (fruit cocktail)	1 cup, diced or sliced, raw or canned
Orange	1 large (3 1/16" diameter)
Peach	1 large (2 ¾" diameter) 2 halves, canned

Food	Serving size
Pear	1 medium pear (2 ½ per lb) 1 cup, sliced or diced, raw cooked, or canned, drained
Pineapple	1 cup, chunks, sliced or crushed, raw, cooked or canned, drained
Plum	1 cup, sliced raw or cooked 3 medium or 2 large plums
Strawberries	About 8 large berries 1 cup, whole, halved, or sliced, fresh or frozen
Watermelon	1 cup, diced or balls
Dried fruit (raisins, prunes, apricots, etc.)	½ cup dried fruit

Cereals and bread group

Cereal group



Rice



Bread



Chapatis

Food	Serving size
Breads	1 regular slice 1 small slice, French 4 snack-size slices rye bread
Cornbread	1 small piece (2 ½" x 1 ¼" x 1¼")
Crackers	5 whole wheat crackers
English muffins	½ muffin
Muffins	1 small (2 ½" diameter)
Oatmeal	½ cup, cooked

Food	Serving size
Pancakes	1 pancake (4 ½" diameter) 2 small pancakes (3" diameter)
Popcorn	3 cups, popped
Ready-to eat cereal	1 cup, flakes or rounds 1 ¼ cup, puffed
Rice	½ cup cooked 1 ounce, dry
Pasta--spaghetti, macaroni,	½ cup, cooked
Tortillas	1 small flour tortilla (6" diameter) 1 corn tortilla (6" diameter)

Fat and oil group



Fat and oil group

Food	Serving size
Vegetable oils (such as canola, corn, cottonseed, olive, peanut, safflower, soybean, and sunflower)	1 Tbsp
Margarine, soft (trans fat free)	1 Tbsp.
Mayonnaise	1 Tbsp
Olives*, ripe, canned	4 large
Avocado*	½ med
Peanut butter*	2 T
Peanuts, dry roasted*	1 oz
Mixed nuts, dry roasted*	1 oz
Cashews, dry roasted*	1 oz
Almonds, dry roasted*	1 oz
Sunflower seeds*	1 oz

Guidelines for using basic food group

- 1- Include at least one or a minimum number of serving from each food group in each meal .
- 2- Make choices with each group as foods within each group are similar but not identical in nutritive value.
- 3- If the meal is vegetarian supplement vegetable proteins with suitable combinations to improve the overall protein quality of the diet. For example (cereal, with small quantities of milk).
- 4- Include uncooked vegetables and fruits in the meals.

Guidelines for using basic food group

- 5- Include at least one serving of milk to ensure a supply of calcium and other nutrients as milk contains all nutrients except iron, vitamin C, and fiber.
- 6- Cereals should not supply more than 75% of total calories.

LECTURE 6

LECTURE 6

Dietary Diversity

Types of pyramids

My Pyramid Basics and Dietary Guidelines



MyPyramid.gov
STEPS TO A HEALTHIER YOU

Dietary Diversity Types of pyramids

1- American Pyramid

2- My Pyramid

3-Vegetarian Pyramid

1-USA OLD

Fats, Oils & Sweets
USE SPARINGLY

KEY

■ Fat (naturally occurring and added)

▼ Sugars (added)

These symbols show fats and added sugars in foods.

**Milk, Yogurt &
Cheese Group**
2-3 SERVINGS



**Meat, Poultry, Fish, Dry Beans,
Eggs & Nuts Group**
2-3 SERVINGS



Vegetable Group
3-5 SERVINGS



Fruit Group
2-4 SERVINGS



**Bread, Cereal,
Rice & Pasta
Group**
**6-11
SERVINGS**



Definition of The food guide pyramid

The food guide pyramid divides food groups into different sections.

- **At the tip of the pyramid (fats, oils, and sugars) indicating limited consumption .**
- **Whereas at the bottom bread, cereals and pastas are, indicating the maximum number of servings .**

The recommended daily serving size

1- Grain and grain products : 6 to 11 serving

2- Vegetables : 3 to 5 serving

3- Fruits : 2 to 4 serving

4- Dairy products : 2 to 3 serving

5- Meat and meat alternatives : 2 to 3 serving

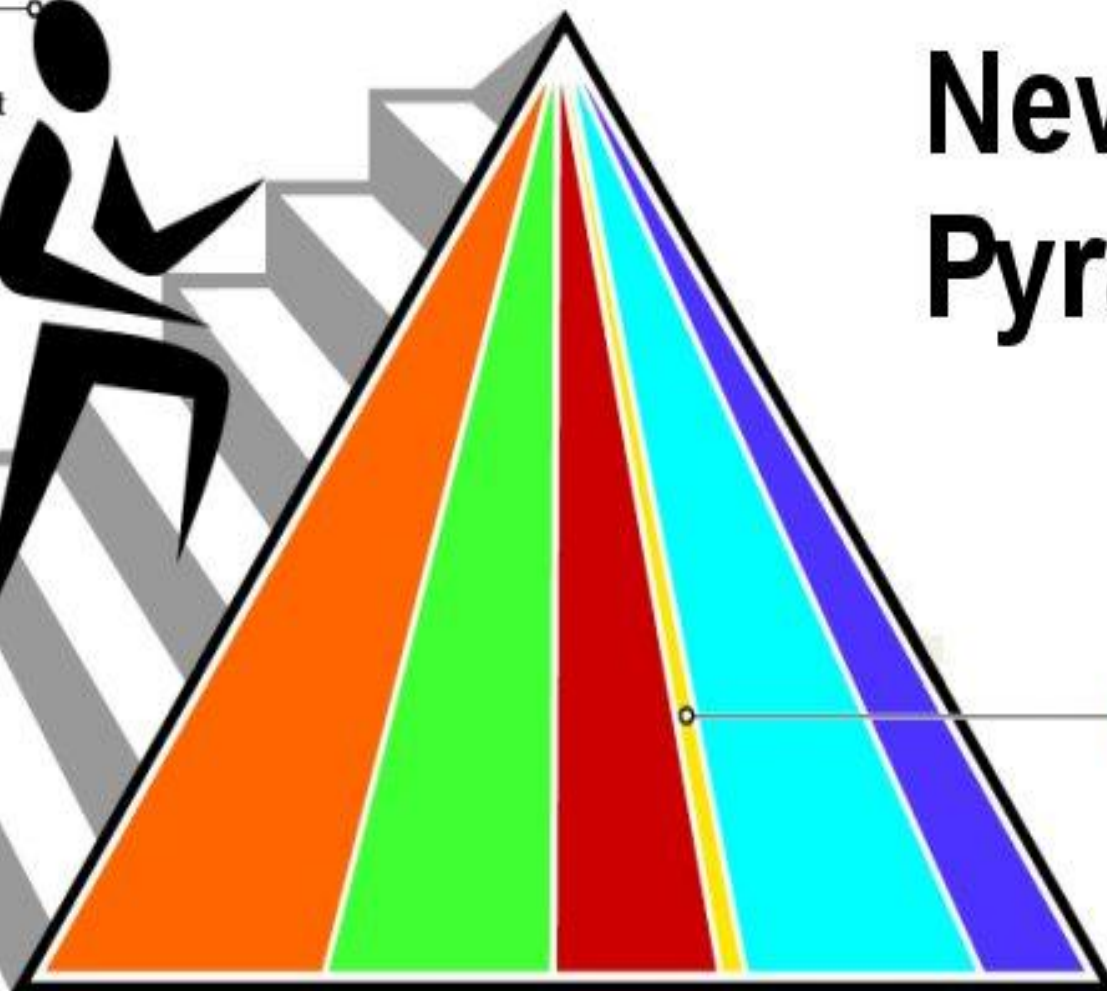
Exercise

- Adults should be physically active for at least 30 minutes most days of the week, children for 60 minutes.
- Sixty to 90 minutes of daily physical activity may be needed to prevent weight gain or sustain weight loss.



New Food Pyramid

2005



Oils

- Most fat should be from fish, nuts and vegetable oils.
- Limit solid fats, such as butter, margarine or lard.
- Keep consumption of saturated fats, trans fats and sodium low.
- Choose foods low in added sugar.

CATEGORY

Grains

Vegetables

Fruits

Milk

Meat and beans

RECOMMENDATION

Half of all grains consumed should be whole grains.

Vary the types of vegetables you eat.

Eat a variety of fruits. Go easy on juices.

Eat low-fat or fat-free dairy products.

Eat lean cuts, seafood and beans. Avoid frying.

DAILY AMOUNT

6 oz.

2.5 cups

2 cups

3 cups

5.5 oz.

Based on a 2,000 calorie diet.

Recommended nutrient intakes at 12-calorie levels can be found on mypyramid.gov.

Anatomy of My Pyramid

Activity

Activity is represented by the steps and the person climbing them, as a reminder of the importance of daily physical activity.

Moderation

Moderation is represented by the narrowing of each food group from bottom to top.

Personalization

Personalization is shown by the person on the steps.

Anatomy of My Pyramid

Proportionality

Proportionality is shown by the different widths of the food group bands.

Variety

Variety is symbolized by the 6 color bands representing the 5 food groups of the pyramid.

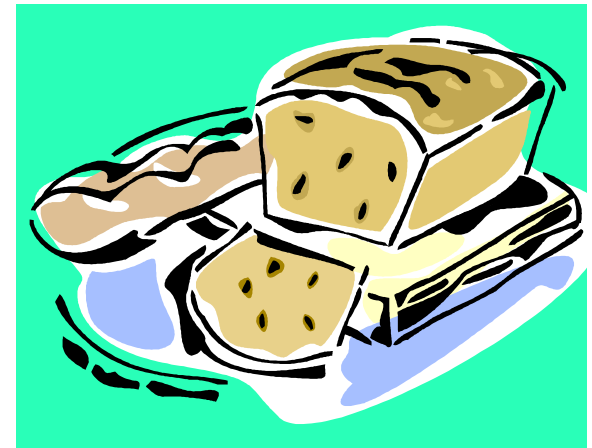
Gradual Improvement

It suggests that individuals can benefit from taking small steps to improve their diet and lifestyle each day.

Grains Group



- Servings Daily: 6 ounces
- Major Nutrient: Carbohydrates, Fiber
- Serving: 1 oz = 1 slice bread – 1 cup dry cereal = $\frac{1}{2}$ cup pasta or rice
- Tips: eat at least 3 oz of whole grains each day



Vegetables Group



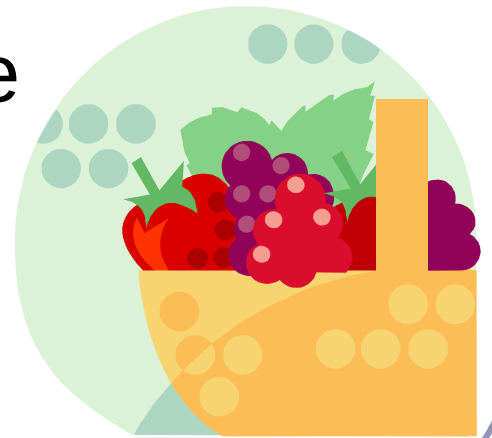
- Servings Daily: 2.5 cups
- Major Nutrient: carbohydrates, fiber, vitamins A and C, folate, potassium, and magnesium
- Serving: Eat a variety of colors
- Tips: Eat more dark green and orange vegetables, eat more dry beans and peas



Fruits Group



- Servings Daily: 2 cups
- Major Nutrient: carbohydrates, fiber, vitamins A and C, folate, potassium, and magnesium
- Serving: 1 medium/small piece
- of fruit = 1 cup
- Tips: eat a variety of fruit, go easy on juices



Milk Group



- Servings Daily : 3 cups
- Major Nutrient: Protein, Minerals, calcium, riboflavin, protein, phosphorus, vitamin B-12
- Serving 1 oz cheese – 1 cup milk/yogurt
- Tips: go low-fat or fat free, if you can't consume milk, choose other calcium sources



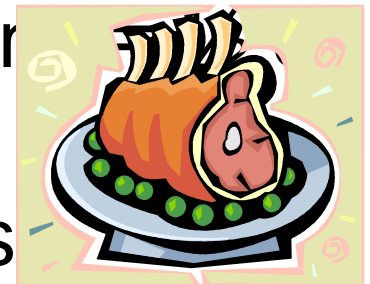
Meat and Beans Group



- Servings Daily: 5 ½ ounces
- Major Nutrient: protein, phosphorus, vitamins B6 and B12, magnesium, iron, zinc, B1, B2, and B3

Serving: 1 oz (30g) meat = 1 egg = 1 T peanut butter = ¼ cup cooked beans or nuts or seeds

- Tips: choose low fat or lean meats
- , bake, broil or grill. Vary protein.



Oils Group

- Servings Daily: 3-11 teaspoons
- Major Nutrient: Fat
- Serving: Watch for it in foods such as nuts, olives, mayonnaise, salad dressing

Tips: Use canola, olive, peanut, soybean, corn safflower or sunflower oil.



Fats and Oils

- Fats are solid at room temperature and oils are liquid.
- Consume less than 10% of calories from saturated fatty acids and less than 300 mg of cholesterol. Keep trans fatty acid consumption as low as possible.
- Choose meat and dairy products that are low in fat.

Physical Exercise



- Be physically active for at least 30 minutes a day.
- About 60 minutes a day of physical activity may be needed to prevent weight gain.
- For sustained weight loss at least 60-90 minutes of physical activity is needed.
- Children and teenagers should be physically active for 60 minutes every day!



Facts on food guide pyramid

1- No one food group is more important than any other regardless of the quantities or servings recommended and that .



Facts on food guide pyramid

2- Foods from one group can not be substituted for foods from another group .

Food items are selected on the basis of nutritional requirements and thus are not substitutable .

For example the meat provides proteins ,and the milk provides calcium .

Food guide pyramid

(Daily recommended food servings according to calories)

Calories (Kcal) Exchanges	1000	1200	1400	1600	1800	2000	2200	2400	2600	2800	3000	3200
Milk	2	2	3	3	3	3	3	3	3	3	4	4
Vegetables	2	2	3	4	4	4	5	5	6	6	6	6
Fruits	3	3	4	4	4	4	4	4	5	5	6	7
Breads & cereals& starch	4	5	6	6	7	7	10	12	13	14	15	16
Meats	4	5	5	6	7	7	7	7	7	8	8	8
Fats	4	5	6	8	8	9	9	10	10	11	12	13

Daily Recommended Food Servings According to Age Groups

Food Group Age Group	Cereal & bread*	Vegetables & fruits	Milk and products	Meat & substitute	Sugar & Fat Minimal amount
Children 2-3 years	3	4	2	1	Sugar & Fat Minimal amount
Children 4-8 years	4	5	2	1	
Children 9- 13 years	6	6	3- 4	1 - 2	
Adolescents 14-18 years females	6	7	3 - 4	2	
Adolescents 14-18 years males	7	8	3 - 4	3	
Adults 19- 50 years females	6	7 - 8	2	2	
Adults 19- 50 years males	8	8 - 10	2	3	
Adults > 50 years females	6	7	3	2	
Adults > 50 years males	7	7	3	3	
Pregnant women	8	8 - 10	4	3	
Lactating mother	8	8 - 10	4	3	

The healthy Food Palm



The Vegan Food Pyramid

Vegetable Oils and Fats
Some sweets, salt,
spices, Nuts.

Use Sparingly

Fortified Dairy Substitutes
2-3 Servings

Eat Moderately

Legumes, Seeds
Beans Group
2-3 Servings

Eat Moderately

Whole Grains, Bread
Rice and Pasta, Cereal Group
6-11 Servings

Eat Generously

VeganFoodPyramid.com

The Veggie Group
3-5 Servings

Eat Liberally

The Fruit Group
2-4 Servings

Eat Liberally

Water

8-10 Glasses a day. If you are active, drink more!

Types of vegetarian menus

1- semi vegetarian : the menu including .

(poultry, fish, eggs, and dairy foods) .

2- ovo- vegetarian : menu including .

(Include eggs in diet).

3- Lacto – vegetarian : menu including .

(Include dairy products, such as cheese, milk, and yogurt, in diet) .

4- Vegan : menu which exclude animal derived foods of all types.

TRADITIONAL CLASSIFICATION OF VEGETARIAN DIETS

Classification	Foods Included	Foods Excluded
Lacto-ovo vegetarian	Fruits, grains, legumes, nuts, seeds, vegetables, milk, milk products, eggs	Meat, poultry, fish, seafood
Lacto-vegetarian	Fruits, grains, legumes, nuts, seeds, vegetables, milk, milk products	Meat, poultry, fish, seafood, eggs
Ovo-vegetarian	Fruits, grains, legumes, nuts, seeds, vegetables, eggs	Meat, poultry, fish, seafood, milk, milk products
Vegan	Fruits, grains, legumes, nuts, seeds, vegetables	Meat, poultry, fish, seafood, eggs, milk, milk products, honey
Macrobiotic	Unrefined/unprocessed grains, commonly brown rice, with smaller amounts of fruits, vegetables and legumes; milk and milk products optional	Meat, poultry, fish (by some), eggs, processed foods, <i>Solanace</i> species (i.e., tomatoes, eggplants and potatoes), supplements (by some)
Fruitarians	Raw fruits, nuts, seeds, berries	All other foods

Advantages of a Vegetarianism

- 1-Vegetarians have lower reported Body Mass Indices**
- 2-* Vegetarians reportedly die less often from heart disease than non-vegetarians**
- 3-Vegetarians Show Lower blood pressure**
- 4-Vegetarians Show Lower blood cholesterol**
- 5-Vegetarians Show Lower prostate and colon cancer of Lower rates**

Disadvantages of Vegetarian Diets

1-Vitamin B12 Deficiency

2-Iron Deficiency

3-Protein Deficiency

Disadvantages of Vegetarian Diets

1- Vitamin B12 Deficiency

**The Vegan Diet does not provide adequate levels
the nutritional needs of vitamin B12 to meet
However through proper person. of the
of vitamin B12 or by consuming supplementation
with vitamin B12, nutritional foods fortified
needs can be met.**

Disadvantages of Vegetarian Diets

2-Iron Deficiency

There are two forms; Heme, which is found only in meat, Non-heme which is found in both meat and plant foods

-Non-heme iron does not absorb as well into the blood, which could cause iron deficiencies in vegans if they are not careful.

***** Good sources of iron include:**

1- Dried fruits

2-Green leafy vegetables

3-Nuts

4-Seeds

5-Whole grains

Disadvantages of Vegetarian Diets

3- Protein Deficiency

- * Eating a variety of unrefined grains, legumes, seeds, nuts, and vegetables throughout the day will provide the recommended daily intake .**
- * Protein combining is not necessary, if a variety of foods are eaten then they will make up the deficit if another food is low in a certain amino acid**

Lecture 7



The Healthy Food Palm For Kingdom of Saudi Arabia



Text book



Physical activity: of 30-60 minutes a day depending on health status



Topics

- **Abbreviations**
- **Definitions**
- **Common diet-related diseases in KSA:**
- **The Dietary Guidelines for Saudis, 2012**
- **The uses of the healthy food palm**

Topics

- **The reasons for choosing the design (The healthy food palm) for the Kingdom of Saudi Arabia:**
- **The palm graphic was chosen due to a number of reasons.**
- **Daily recommended Food serving according to age groups**
- **Rations allowable daily food groups**
- **In The Healthy Food Palm**

Daily Recommended Food Servings According to Age Groups

Food Group Age Group	Cereal & bread*	Vegetables & fruits	Milk and products	Meat & substitute	Sugar & Fat Minimal amount
Children 2-3 years	3	4	2	1	Sugar & Fat Minimal amount
Children 4-8 years	4	5	2	1	
Children 9- 13 years	6	6	3- 4	1 - 2	
Adolescents 14-18 years females	6	7	3 - 4	2	
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Adults > 50 years females	6	7	3	2	
Adults > 50 years males	7	7	3	3	
Pregnant women	8	8 - 10	4	3	
Lactating mother	8	8 - 10	4	3	

The healthy Food Palm



Exchange lists for meal planning
-Serving size for groups

Exchange lists for meal planning

Exchanges	No . Of units	Carbohydrate	Protein	Fats	Calories
Milk					
a-Skim milk	1	12	8	-	80
b-Low fat		12	8	5	125
c- high fat milk		12	8	10	170
Non-starchy Vegetables	1	5	2	-	25
Fruits	1	15	-	-	60
Breads & cereals& starch vegetables	1	15	2	-	68
Meats					
a-Very low fat	1	-	7	0-1	35
b-Lean fat meat		-	7	3	55
b-Medium fat meat		-	7	5	73
c-High fat meat		-	7	8	100
Eats	1	-	-	5	45

Exchanges	Size of exchange	Importance of group
Milk	1 cup milk, 1 Yoghurt, 1 cup Fermented milk, , 10 gm Cow milk powder, 30gm Cream	1) *Dairy foods are good sources of calcium, riboflavin, protein, phosphorus, vitamin B-12 2) Many are fortified with vitamins A and D 3) Suggest low-fat or fat-free dairy products 4) Daily serving/day; 2-4
Vegetables	1 medium , or 1 cup fresh, or 1/2 cup cooked of (Tomato, Carrot, Cucumber, Lettuce, Cabbage, Cauliflower, Onion , Garlic, Parsley, Coriander, Celery, watercress, Garden rocket, Green pepper, Radish , Leek bulbs, Olive black) 1 medium , or 1 cup fresh, or 1/2 cup cooked of (Okra, Squash, Pumpkin, Spinach, Jaw's mallow, eggplant) Green fenugreek	1) Vegetables are good sources of carbohydrates, fiber, vitamins A and C, folate, potassium, and magnesium 2) Eat more dark green vegetables 3) Eat more orange vegetables 4) Eat more dry beans and peas 5) Daily serving/day; 3-5

Exchanges	Size of exchange	Importance of group
Fruits	❖ 1/2 cup , or medium size, (Orange, Apple, Apricot, Mandarine, ½ large Banana, Cantalop, Pear,, Fig, peach,, grapefruit, Guava ❖ 3 Dates dried ❖ 17 Graps ❖ 4 Apricots ❖ 12 cherries ❖ 2 tbsp dried fruits ❖ 2 medium figs ❖ ½ cup fruit cocktail ❖ 2 plums ❖ 3 Prunes ❖ 2 TPsp raisins ❖ 1 ¼ cup watermelon ❖ 1/2 cup , or medium size, (Plum, Watermelon, Mango, Melon sweet, pineapple, Strawberry, Lemon, Lime, Mulberry, persimmon, Pomegranate, Raisins, raspberry,	1)Fruits are good sources of carbohydrates, fiber, vitamins A and C, folate, potassium, and magnesium 2)Eat a variety of fruit 3)Choose fresh, frozen, canned or dried fruit 4)Go easy on fruit juices Daily serving/day; 2-4

Guidelines for Exchange lists

Exchanges	Size of exchange	Importance of group
Breads & cereals& starch	1/4 Balady bread, ½ French bread 1/4 White bread, 1 slice Toast ½Corn bread, 3 Biscuit 1 Fattayer with zatar cracker cookies cakes Pancake ½ cup Rice (brown, white) ½ cup Macaroni ½ cup Wheat Wheat parboiled ½ cup Barly Wheat flour 72% extraction Colocasia tube Potato white Sweat potato bagel nasta	1) *Eat at least 3 ounces of whole grain breads, cereal, crackers, rice, or pasta each day 2) *They are rich fiber 3) *They are good sources of the folate, riboflavin, zinc, protein, and magnesium thiamin, niacin, iron, Daily serving/day; 6-11

Exchanges	Size of exchange	Importance of group
Meats	❖ 30 gm meat (Lamb, Beef, Buffalo, Goat, Canned corned beef, Camel, turkey, veal) ❖ 30 gm fish ❖ 1/4 (Chicken, Poultry) ❖ 1 egg ❖ 30 gm (Cheese camembert, Cheese, cheddar, Cream, Whole milk cheese ❖ Skim cheese) ❖ 30 gm (Trianglecheese, Kiri cheese ❖ Sliced cheese, Cheese roquefort) ❖ Dried meat (basterma) ❖ Heart, Kidney, liver ❖ Pigon ❖ Rabbit ❖ Salami ❖ Sausage, beef ❖ soyabean ❖ Dry beans ❖ Cowpea Peas garden Lentils French common beans	1) *Include meat, poultry, fish, dry beans, eggs, nuts Encourage low-fat or lean meats and poultry Cooking methods: baking, broiling, grilling 2)Good sources of protein, phosphorus, vitaminsB6and B12, magnesium, iron, zinc, B1, B2, and B3 3)Legumes: good sources of fiber and vitamins (vegetables), proteins and minerals (meats) Daily serving/day; 2-3

Guidelines for Exchange lists

Exchanges	Size of exchange	Importance of group
Fats	Monounsaturated fat : 1 tea spoon , 5 gm or 6 nuts (Almonds , Cashews, Hazel nuts, Pea nut, nut, mixed) 16 nuts Pistachio 10 nuts peanuts 8 larges olives 1 tsp (olive oil, canola oil, peanut oil) Polyunsaturated fats (1 Tbsp margarine reduced fat, 1 Tpsp pignolia (pine nuts) 4 halves Walnuts, 1tsp corn oil, grape seed, soybean oil, sunflower oil, 2 Tpsp salad dressing reduced fat, 1 Tbsp sesame oil, 2 tsp Halawah Tahini, sesame paste Saturated fats 1 Tbsp butter, 2 Tbsp cream, 1 ½ Tbsp cream cheese, 1 tsp coconut oil, 1 tsp palm oil, 1 tea spoon , 5 gm Chocholata	1) Encourage people to select health-promoting 2) forms of fats: fat from fish, nuts, and vegetable oils 3) Limit solid fats: butter, stick margarine, shortening, lard, and visible fat on meat 4) Daily serving/day; 2-3